



Aakash

Medical | IIT-JEE | Foundations

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MM : 720

Zoology 12th

Time : 180 Min.

Zoology

- Nearly all fossils occur in what kind of rocks?
 - (1) Batholithic
 - (2) Igneous
 - (3) Metamorphic
 - (4) Sedimentary
- Match the following w.r.t. convergent evolution.

a. Bobcat	(i) Tasmanian wolf
b. Wolf	(ii) Tasmanian tiger cat
c. Anteater	(iii) Marsupial mole
d. Mole	(iv) Numbat

 Select the **correct** option.
 - (1) a(i), b(ii), c(iii), d(iv)
 - (2) a(iv), b(i), c(ii), d(iii)
 - (3) a(ii), b(iv), c(i), d(iii)
 - (4) a(ii), b(i), c(iv), d(iii)
- Select the **correct** match.
 - (1) Syphilis – *Haemophilus ducreii*
 - (2) Genital warts – *Treponema pallidum*
 - (3) AIDS – Human Papilloma Virus
 - (4) Malaria – *Plasmodium vivax*
- Match the following.

	Column I		Column II
a.	Theory of special creation	(i)	Conjectured that first forms of life came from pre existing non-living organic molecules
b.	Cosmozoic theory	(ii)	Units of life called spores were transferred from different planets to earth.
c.	Theory of chemical evolution	(iii)	All living organisms that we see today were created as such
d.	Big Bang theory	(iv)	Explains the origin of universe

 - (1) a(iii), b(ii), c(i), d(iv)
 - (2) a(iii), b(iv), c(i), d(ii)
 - (3) a(ii), b(i), c(iv), d(iii)
 - (4) a(i), b(ii), c(iv), d(iii)
- Presence of coronavirus, if present even in little amount can be traced using
 - (1) RT-PCR
 - (2) Urine analysis
 - (3) Stool analysis
 - (4) Widal test
- Taq* polymerase enzyme is obtained from
 - (1) *Bacillus thuringiensis*
 - (2) *Thermus aquaticus*
 - (3) *Clostridium botulinum*
 - (4) *Pseudomonas putida*
- Read the below given features about drug 'X'.
 - (a) Obtained from plant native to South America
 - (b) Interferes with the transport of the neurotransmitter dopamine
 - (c) Has a potent stimulating action on central nervous system.
 Identify 'X' and select the **correct** option.
 - (1) Cocaine
 - (2) Cannabinoid
 - (3) Morphine
 - (4) Smack
- Assertion (A)** : A patient with ADA deficiency requires periodic infusion of own genetically engineered lymphocytes containing functional ADA cDNA.
Reason (R) : The lymphocytes have a limited life span and their death causes enzyme deficiency again in the patient.
 In the light of above statements select the **correct** option.
 - (1) Both Assertion and Reason are true and Reason is the correct explanation of the Assertion
 - (2) Both Assertion and Reason are true but Reason is not the correct explanation of the Assertion
 - (3) Assertion is true but Reason is false
 - (4) Both Assertion and Reason are false

9. Choose the **incorrect** statement for PCR technique

- (1) This technique can be used for molecular diagnosis of many diseases
- (2) In this technique, thermolabile enzyme Taq-polymerase is used
- (3) The second step of PCR technique is annealing of oligonucleotide primers with its template DNA at 50°-60°C temperature
- (4) Taq polymerase synthesizes DNA in 5' → 3' direction using co-factor Mg^{++} and dNTPs

10. At the time of implantation, 'X' stage of rapidly dividing embryo becomes embedded in the 'Y' layer of uterus. Choose the option which gives the **correct** answer for X and Y in above statement.

- (1) X-Morula, Y-Endometrial
- (2) X-Blastomere, Y-Myometrial
- (3) X-Blastocyst, Y-Endometrial
- (4) X-Morula, Y-Myometrial

11. Which among the following depicts the **correct** direction of flow of milk?

- (1) Mammary alveolus – Mammary duct – Mammary tubules – Mammary ampulla – Lactiferous duct
- (2) Mammary alveolus – Mammary tubules – Mammary duct – Mammary ampulla – Lactiferous duct
- (3) Mammary ampulla – Mammary duct – Mammary tubules – Mammary alveolus – Lactiferous duct
- (4) Mammary alveolus – Mammary duct – Mammary ampulla – Mammary tubules – Lactiferous duct

12. **Assertion (A)** : Before industrialization, there were more white winged moths on trees than melanised moths.

Reason (R) : Dark color in moths evolved due to a mutation induced by chemicals present in industrial smoke.

In the light of above statements, select the **correct** option

- (1) Both Assertion and Reason are true and Reason is the correct explanation of the Assertion
- (2) Both Assertion and Reason are true but Reason is not the correct explanation of the Assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

13. PCR is used for

- (1) Reverse transcribing RNA into DNA
- (2) Amplifying DNA
- (3) Digesting DNA
- (4) Separation of DNA fragments

14. Mark the **correct** option.

- (1) Mucous coating of epithelium lining the GIT is considered as a physiological barrier.
- (2) Saliva in the mouth is an example of physical barrier of innate immunity
- (3) Interferons which protect virus-infected cells are glycoproteins that constitute cytokine barriers
- (4) Macrophages in tissues, natural killer cells, PMNLs and monocytes in blood are part of cellular barriers providing innate immunity

15. A : Blood of mother never mixes with that of foetus during pregnancy.
R : Nutrients and gases are transported by diffusion across placenta.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

16. Choose the **incorrect** match.

- | | |
|-----------------------------------|------------------------|
| (1) Site of fertilisation | – (A) Fallopian tube |
| (2) Temporary endocrine structure | – (B) Corpus luteum |
| (3) Hysterectomy | – (C) Removal of ovary |
| (4) Acellular layer | – (D) Zona pellucida |

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

17. How many of the following statements are **correct** for Saheli?

- (i) Selective estrogen receptor modulator
- (ii) Inhibits implantation
- (iii) Injectable device, developed by CDRI, Lucknow
- (iv) Steroidal contraceptive
- (v) From the beginning taken once in a week

- (1) 5
- (2) 4
- (3) 3
- (4) 2

18. Read the following statement A and B and choose the **correct** option.

A. Adaptive radiation is process of evolution of different species in a given geographical area starting from a point and literally radiating to other areas of geography.

B. Australian marsupial (Bobcat) and placental mammal (Tasmanian tiger cat) are example of divergent evolution.

- (1) Statement A is correct but B is incorrect
- (2) Statement A is incorrect but B is correct
- (3) Both statements are correct
- (4) Both statements are incorrect

19. A : Female foeticide is punishable under prenatal diagnostic technique act.
R : Abortion after illegal sex determination has led to seriously altered sex ratio in many parts of India.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

20. Which of the following is **incorrect** w.r.t different hominids, their cranial capacities and characteristics?

	Hominid	Cranial capacity	Characteristics
(1)	<i>Homo habilis</i>	650-800 cc	Vegetarian, tool-maker
(2)	<i>Homo erectus</i>	900 cc	Discovered fire, ate meat
(3)	Neanderthal man	1400 cc	Used hides as clothing and had burial customs
(4)	<i>Australopithecus africans</i>	1700 cc	Bipedal, used stone tools

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

21. Which of the following immunoglobulins will be produced in an individual in case of an allergy attack?

- (1) Ig G
- (2) Ig M
- (3) Ig E
- (4) Ig A

22. RNAi method involves silencing of a specific _____ to inhibit the process of _____. Choose the option which fill the blanks **correctly**.

- (1) rRNA ; replication
- (2) mRNA; translation
- (3) mRNA ; transcription
- (4) tRNA; translation

23. The protein encoded by gene *cry/Ab* controls

- (1) Silkworm
- (2) Beetles
- (3) Corn borer
- (4) Flies

24. A protoxin is

- (1) A primitive toxin
- (2) An active toxin
- (3) Toxin produced by protozoans only
- (4) Inactive toxin

25. A : Fins of shark and flippers of dolphin show convergent evolution.

R : Proteins found in blood of man and apes are examples of molecular homology.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

26. The contraceptive device inserted by the doctors or expert nurses in the uterus through vagina is

- (1) Implant
- (2) IUD
- (3) Cervical cap
- (4) Diaphragm

27. **Assertion (A):** Inner cell mass carries the potential to differentiate into all the three germ layers.

Reason (R): Inner cell mass contains the stem cells.

In the light of above statements, choose the **correct** answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true and (R) is false
- (4) (A) is false and (R) is true

28. Interstitial spaces in human testis does not contain

- (1) Leydig cells
- (2) Small blood vessels
- (3) Sertoli cells
- (4) Immunologically competent cells

29. Liver cirrhosis is mostly caused by the chronic intake of

- (1) Heroin
- (2) Tobacco
- (3) Morphine
- (4) Alcohol

30. Which of the following is an intratesticular accessory duct?

- (1) Seminiferous tubule
- (2) Rete testis
- (3) Ejaculatory duct
- (4) Vas deferens

31. α -1-antitrypsin is given to cigarette smokers for treatment of a disease called
- (1) Emphysema
 - (2) Asthma
 - (3) Lung carcinoma
 - (4) Lung fibrosis
32. In biotechnology, a selectable marker is used to
- (1) Decrease copy number of plasmid
 - (2) Increase copy number of plasmid
 - (3) Help in elimination of the non-transformants
 - (4) Facilitate autonomous replication of plasmid
33. Industrial melanism is an example of
- (1) Protective mimicry
 - (2) Appearance of darker coloured individual due to very poor sunlight
 - (3) Natural selection in which darker forms were selected
 - (4) Natural selection in which lighter forms were selected
34. Following is a list of some diseases. Find out how many of them are **not** caused by a bacterial pathogen.
- i. Typhoid
 - ii. Common cold
 - iii. AIDS
 - iv. Amoebiasis
 - v. Malaria
 - vi. Yellow fever
 - vii. Ringworm
 - viii. Asthma
- (1) Two
 - (2) Four
 - (3) Seven
 - (4) Eight
35. Histopathological examinations is also known as
- (1) Chemotherapy
 - (2) MRI
 - (3) X-rays
 - (4) Biopsy
36. The first transgenic cow, Rosie, produced human protein-enriched milk in the year 'Q'. Choose the event that also occurred in year 'Q'.
- (1) Production of recombinant form of human insulin.
 - (2) First gene therapy given to a 4-year old girl with ADA deficiency.
 - (3) An American company got patent rights on Basmati rice via US Patent and Trademark Office.
 - (4) Enzymes responsible for restricting the growth of bacteriophage in *E.coli* was isolated.
37. 95% of all existing transgenic animals are
- (1) Rabbits
 - (2) Rats
 - (3) Mice
 - (4) Monkey
38. The technique that involves formation of double stranded RNA in a cell, further inhibiting the translation of pathogen specific mRNA is
- (1) PCR
 - (2) Gene therapy
 - (3) RNA interference
 - (4) Batch culture
39. Bt cotton genes prevent damage to crops from
- (1) Bacterial pathogens
 - (2) Fungal pathogens
 - (3) Nematode parasites
 - (4) Insect pests
40. GM plants have been useful in many ways **except**
- (1) Helps to reduce post harvest losses
 - (2) Enhances nutritional value of food
 - (3) Decreases efficiency of mineral usage by plants
 - (4) Reduced reliance on chemical pesticides
41. Mark the odd one out w.r.t. advantages of amniocentesis.
- (1) Detection of biochemical abnormalities
 - (2) Prenatal diagnostics
 - (3) Detection of enzymatic abnormalities
 - (4) Determination of sex to abort female foetus
42. Which of the following is **not** secreted by placenta?
- (1) hCG
 - (2) hPL
 - (3) Estrogen
 - (4) Cortisol
43. Assertion : Biologically, woman is superior to man.
Reason : A woman provides nutrition to developing foetus during gestation and also nourishes the baby on her milk after its birth.
- (1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion
 - (2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion
 - (3) If the assertion is true but the reason is false
 - (4) If the assertion is false but reason is true

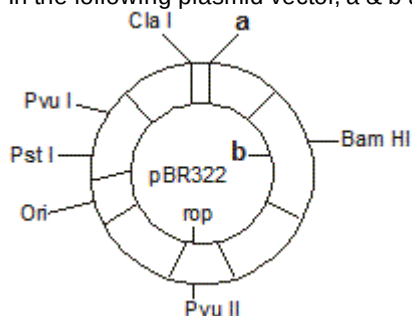
44. The exaggerated response of our immune system to certain antigens present in the environment results in

- (1) Withdrawal syndrome
- (2) Allergic response
- (3) Auto-immune disease
- (4) Amnestic response only

45. Progestasert is a/an

- (1) Popular brand of oral contraceptive pills
- (2) Non-medicated IUD
- (3) Hormone releasing IUD
- (4) Implant containing copper only

46. In the following plasmid vector, a & b are respectively



- (1) a = Hind III, b = tet^R
- (2) a = Hind II, b = amp^R
- (3) a = Sal I, b = tet^R
- (4) a = EcoRI, b = amp^R

47. In human beings, menstrual cycles cease around 50 years of age, this is termed as

- (1) Implantation
- (2) Ovulation
- (3) Menarche
- (4) Menopause

48. Earth was supposed to have been formed about

- (1) 6 bya
- (2) 4.5 bya
- (3) 15 bya
- (4) 2.5 bya

49. Match the diseases listed in Column-I with their symptom/Diagnostic test/Pathophysiology listed in Column-II.

Column-I	Column-II
(a) Malaria	(i) Widal
(b) Typhoid	(ii) Dysentery
(c) Amoebiasis	(iii) Scaly lesions
(d) Ringworm	(iv) Haemozoin

Choose the correct answer from the options given below.

- (1) a-iv, b-i, c-iii, d-ii
- (2) a-iii, b-iv, c-ii, d-i
- (3) a-iv, b-i, c-ii, d-iii
- (4) a-ii, b-iv, c-iii, d-i

50. The first vaccination administered by Edward Jenner was to protect people from

- (1) Polio
- (2) Influenza
- (3) Small pox
- (4) Rabies

51. Cave paintings were done by

- (1) *Homo sapiens neanderthalensis*
- (2) *Homo sapiens fossilis*
- (3) *Homo habilis*
- (4) *Homo erectus*

52. Read the following statements

Statement A : Restriction enzymes are obtained only from eukaryotes.

Statement B : The first discovered restriction endonuclease was *EcoRI*.

Choose the **correct** option.

- (1) Both statements A and B are correct
- (2) Both statements A and B are incorrect
- (3) Statement A is correct but B is incorrect
- (4) Statement B is correct but A is incorrect

53. **Assertion (A)** : Vasectomy has no effect on sexual act of the male.

Reason (R) : He can still capulate, ejaculate and has normal secretion of sex hormones.

In the light of above statements, choose the correct option.

- (1) Both Assertion & Reason are true and Reason is the correct explanation of the Assertion
- (2) Both Assertion & Reason are true but Reason is not the correct explanation of the Assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

54. Diaphragms, cervical caps, and vaults are used to cover

- (1) Vagina
- (2) Cervix
- (3) Penis
- (4) Fundus of uterus

55. Cycads and dicotyledons arose from common ancestor called '**A**' that existed on Earth during period '**B**'. Choose the option which fills the blanks 'A' and 'B' **correctly**.
- (1) *Zosterophyllum* and Triassic
 - (2) *Chlorophyta* and Silurian
 - (3) Seed ferns and Carboniferous
 - (4) Herbaceous lycopods and Cretaceous
56. The first clinical gene therapy was given in year
- (1) 1990
 - (2) 1991
 - (3) 1992
 - (4) 1993
57. A : Innate immunity is a non-specific type of immunity.
R : B-cells are part of innate immunity.
- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
 - (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
 - (3) Assertion is true statement but Reason is false
 - (4) Both Assertion and Reason are false statements
58. Most important event during third week of development is
- (1) Morulation
 - (2) Blastulation
 - (3) Gastrulation
 - (4) Foetal heart sound
59. In the case of SCID, which technique is considered to be completely curative?
- (1) Oral intake of ADA tablets
 - (2) Bone marrow transplantation
 - (3) Enzyme replacement therapy
 - (4) Gene therapy in early embryonic stage
60. dNTP stands for
- (1) Deoxyribonucleotide triphosphate
 - (2) Deoxyribonucleoside triphosphate
 - (3) Deoxyribonucleotide monophosphate
 - (4) Deoxyribonucleoside monophosphate
61. Which of the following statements is **incorrect** regarding test tube baby?
- (1) Early embryo (with upto 8 blastomeres) is transferred into the fallopian tube
 - (2) Embryo with more than 8 blastomeres is transferred into the uterus
 - (3) ICSI method is also considered as a type of IVF
 - (4) Ovum gets implanted in the fallopian tube and further development occurs in uterus
62. By the end of A of human pregnancy, the foetus develops limbs and digits and by the end of B most of the major organ systems are formed. Choose the option that fill the blanks **correctly**.
- (1) **A**– First month , **B**– First trimester
 - (2) **A**– Second month , **B**– Second trimester
 - (3) **A**– Second month, **B**– First trimester
 - (4) **A**– First month, **B**– Third trimester
63. After the first trimester, which of the following steroid hormones is/are produced by placenta?
- (1) Estrogen only
 - (2) Estrogen, progesterone
 - (3) Estrogen, progesterone, hCG
 - (4) Estrogen, progesterone, hCS
64. On what days of menstrual cycle, a couple should abstain from coitus in order to prevent pregnancy, if the menstrual cycle of female is of 34 days?
- (1) Day 10 to 17
 - (2) Day 24 to 30
 - (3) Day 27 to 34
 - (4) Day 16 to 23
65. During detection of cancer, technique that uses ionising rays to generate a three dimensional image of internal organs of a patient is
- (1) Autoradiography
 - (2) CT-scan
 - (3) MRI
 - (4) Biopsy
66. To clone a gene of interest from genomic DNA of tomato by PCR, we require
- a. Primers
 - b. dATP
 - c. dUTP
 - d. *Taq* polymerase
 - e. ATP
- Select the **correct** answer from the options given below.
- (1) a, b, c, d
 - (2) b, c, d, e
 - (3) Only a and d
 - (4) Only a, b and d
67. Which of the following is **not** a steroidal hormone released during pregnancy?
- (1) Progesterone
 - (2) Estradiol
 - (3) hCG
 - (4) Cortisol

68. Choose the **incorrect** statement.

- (1) Smallest DNA molecules move the fastest in gel electrophoresis
- (2) *Bam* HI occupies a restriction site in ampicillin resistance gene of pBR322
- (3) A piece of DNA may be introduced in a host bacterium through transformation
- (4) 'rop' codes for proteins involved in replication of plasmid

69. Choose the correct sequence of embryonic development in humans

- (1) Zygote → Morula → Gastrula → Blastocyst
- (2) Zygote → Morula → Blastocyst → Gastrula
- (3) Gastrula → Morula → Zygote → Blastocyst
- (4) Zygote → Gastrula → Morula → Blastocyst

70. Mark the event which occurs under the influence of parasympathetic impulse?

- (1) Emission
- (2) Ejaculation
- (3) Erection of penis
- (4) Closing of urethral sphincters

71. **Assertion** : For isolation of DNA from tomato, the enzyme cellulase is frequently used.

Reason : It is a protease that helps to digest cellulose in plant cell wall.

In the light of above statements, select the **correct** option.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

72. Choose the **incorrect** statement w.r.t. AIDS.

- (1) It is caused by HIV which is termed retrovirus as it has a RNA genome
- (2) Latent phase of the disease can last from 6 months to 10 years in AIDS patient
- (3) Macrophages are called factories for HIV multiplication in the body
- (4) Many opportunistic infection such as Toxoplasmosis is caused by protozoan *T.gondii*

73. In 1990, a four year old girl suffering from SCID was given _____ to overcome the deficiency. Choose the option that fills the blank **correctly**.

- (1) Chemotherapy
- (2) Gene therapy
- (3) Radiotherapy
- (4) Physiotherapy

74. A : A patient with ADA deficiency requires periodic infusion of own genetically engineered lymphocytes containing functional ADA cDNA.

R : The lymphocytes have a limited life span and their death again restores condition of enzyme deficiency in the patient.

In the light of above statements, select the correct option.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

75. C-peptide of human insulin is

- (1) A part of mature insulin molecule
- (2) Responsible for formation of disulphide bridges in mature insulin
- (3) Removed during maturation of proinsulin to insulin
- (4) Responsible for biological activity of humulin

76. Which of the following represents **correct** sequence of evolution of mammals from reptiles?

- (1) Early reptiles → Sauropsids → Thecodonts → Mammals
- (2) Synapsids → Pelycosaur → Therapsids → Mammals
- (3) Thecodonts → Therapsids → Dinosaurs → Mammals
- (4) Early reptiles → Thecodonts → Therapsids → Mammals

77. A : Spleen is considered as the graveyard of dead RBC's. R : The RBC's and other blood cells are eaten by majorly by neutrophils in the spleen.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

78. **Assertion (A)** : Industrial melanism illustrates the role of anthropogenic action in evolutionary process.

Reason (R) : Pollution generated by human activity led to origin of darker variety of moths.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

79. Who among the following buried their dead and used hides to protect their body?

- (1) *Homo erectus*
- (2) *Homo habilis*
- (3) *Homo sapiens neanderthalensis*
- (4) *Ramapithecus*

80. Match the following columns and select the **correct** option.

	Column I		Column II
a.	Primary oocyte	(i)	Arrested at prophase - I
b.	Zona pellucida	(ii)	Granulosa cells
c.	Corona radiata	(iii)	Glycoproteinaceous acellular layer
d.	Secondary oocyte	(iv)	Arrested at metaphase-II

- (1) a(iv), b(iii), c(ii), d(i)
 (2) a(i), b(ii), c(iii), d(iv)
 (3) a(iii), b(ii), c(i), d(iv)
 (4) a(i), b(iii), c(ii), d(iv)
81. A : Hugo de Vries said that evolution is a slow and gradual process.
 R : According to him, variations are non-random and directional.
 (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion.
 (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion.
 (3) Assertion is true statement but Reason is false.
 (4) Both Assertion and Reason are false statements.
82. A : In gene therapy, a normal allele is cloned and the DNA is introduced into certain human cells where its expression may be sufficient to yield a normal phenotype.
 R : A technical challenge in gene therapy is finding safe and effective vector to deliver the gene of interest into the cells.
 (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
 (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
 (3) Assertion is true statement but Reason is false
 (4) Both Assertion and Reason are false statements
83. "Sparrows with average sized wings survive strong wind area better than those with longer or shorter wings." What type of selection it is?
 (1) Stabilising selection
 (2) Balancing selection
 (3) Diversifying selection
 (4) Directional selection
84. Amniocentesis is a foetal sex determination test based on the chromosomal pattern in cell from
 (1) Chorionic villi
 (2) Umbilical cord
 (3) Amniotic fluid
 (4) Bone marrow

85. Choose the **odd** one w.r.t. ploidy.

- (1) Spermatogonia
 (2) Primary spermatocyte
 (3) Primary oocyte
 (4) Secondary oocyte

86. Which of the following contraceptive is an injectable form of the birth control device that prevents ovulation?

- (1) Norplant
 (2) Depo-Provera
 (3) Saheli
 (4) Rhythm method

87. A : Development of similar traits may occur in distantly related species living in similar habitats.
 R : These similarities in distantly related species develop as a result of divergent evolution.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
 (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
 (3) Assertion is true statement but Reason is false
 (4) Both Assertion and Reason are false statements

88. Match column I with column II and select the **correct** option.

	Column I		Column II
a.	Fimbriae	(i)	Columnar ciliated epithelium
b.	Fallopian tube	(ii)	Capture ova released into coelom
c.	Infundibulum	(iii)	Site of fertilization
d.	Ampulla	(iv)	Part of oviduct closer to ovary

- (1) a(iv), b(i), c(ii), d(iii)
 (2) a(ii), b(i), c(iv), d(iii)
 (3) a(i), b(ii), c(iii), d(iv)
 (4) a(i), b(iii), c(iv), d(ii)

89. A : In a malaria infected person, chill and high fever recur every 3-4 days.

R : Parasite infected RBCs get ruptured, thereby releasing a toxic substance haemozoin.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
 (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
 (3) Assertion is true statement but Reason is false
 (4) Both Assertion and Reason are false statements

90. A : Usually sperm and ovum of two different species are unable to fuse together.
R : Fertilizin of egg interacts with the antifertilizin of sperm of same species only.
- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
 - (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
 - (3) Assertion is true statement but Reason is false
 - (4) Both Assertion and Reason are false statements
91. Chitinase acts on
- (1) Bacteria
 - (2) Animal cell
 - (3) Plant cell
 - (4) Fungus
92. Out of 200-300 million sperms in ejaculate of a human, at least how many sperms must have normal shape and size and must also show vigorous motility for normal fertility?
- (1) 120-180 million
 - (2) 48-72 million
 - (3) 100-150 million
 - (4) 150-200 million
93. A : Coacervates were probably non cellular form of life that could have originated during evolution.
R : Coacervates were the self duplicating aggregates of proteins surrounded by lipid layer.
- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
 - (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
 - (3) Assertion is true statement but Reason is false
 - (4) Both Assertion and Reason are false statements
94. Data from experimental survey in India suggests that, weight of new borns, ranges between 2 kg to 5 kg but majority of surviving new borns are around 3 kg in weight. Which of the following is **correct** for such type of natural selection when the graph is plotted between number of surviving babies and their weight?
- (1) It is a type of directional selection
 - (2) The peak formed in graph corresponds to mean character value
 - (3) The peak formed in graph corresponds to peripheral character values
 - (4) Two peaks are obtained corresponding to extreme character values in the graph
95. Choose the **incorrect** match w.r.t. insertion of rDNA into host cell.
- (1) Electroporation – Transient microscopic pores in membrane
 - (2) Biolistics – Gene gun method
 - (3) Transformation – Heat shock method
 - (4) Microinjection – CaCl_2 is injected into nucleus
- (1) (1)
 - (2) (2)
 - (3) (3)
 - (4) (4)
96. A genetically modified brinjal in India was developed with an insert of *cryIAb* gene. This was done to develop resistance to
- (1) Low mineral content of soil
 - (2) Drought and disease
 - (3) Insects
 - (4) Softening of fruit
97. Human immunodeficiency virus primarily infects
- (1) Eosinophils
 - (2) T_4 lymphocytes
 - (3) Cytotoxic T cells
 - (4) Erythrocytes
98. Downstream processing involves
- (1) Obtaining the foreign gene product from host cell but not storage and selling
 - (2) Culturing host cell in bioreactors
 - (3) Amplification of gene of interest
 - (4) Purification and separation of product
99. Read the given statements.
- Statement I :** Restrictions imposed for MTP promote indiscriminate and legalised female foeticide in India.
- Statement II :** MTP is most widely accepted method of conception in our country.
- Select the **correct** option.
- (1) Only statement I is true
 - (2) Only statement II is true
 - (3) Both statements I and II are true
 - (4) Both statements I and II are false

100. Match column I and column II and select the **correct** option using the codes given below.

	Column I		Column II
a.	NK cell	(i)	Basophil
b.	Complement system	(ii)	Neutrophil
c.	Mast cell	(iii)	Innate immunity
d.	PMNL	(iv)	Blood proteins

(1) a(iii), b(iv), c(ii), d(i)

(2) a(iii), b(iv), c(i), d(ii)

(3) a(iii), b(ii), c(i), d(iv)

(4) a(i), b(ii), c(iii), d(iv)

101. A : Natural killer cells destroy infected body cells and cancer cells by phagocytosis.

R : Natural killer cells is one of the body's most potent defences.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

102. The first hormone to peak in secretory phase of menstrual cycle is 'A' while decline in 'B' leads to regression of corpus luteum.

Select the option that fill the blanks **correctly**.

	A	B
(1)	Estrogen	FSH
(2)	Progesterone	LH
(3)	LH	Progesterone
(4)	Progesterone	Estrogen

(1) (1)

(2) (2)

(3) (3)

(4) (4)

103. Foetal ejection reflex works through

(1) Positive feedback exerted by progesterone on hypothalamus

(2) Positive feedback exerted by estrogen on endometrium

(3) Positive feedback exerted by oxytocin on hypothalamus

(4) Negative feedback exerted by oxytocin on hypothalamus

104. Nematode which infects the roots of tobacco plants and causes a great reduction in yield is

(1) *Bacillus thuringiensis*

(2) *Meloidogyne incognita*

(3) *E. coli*

(4) *Agrobacterium tumefaciens*

105. Match the following fossils of man w.r.t. their cranial capacity.

	Column I		Column II
a.	<i>Homo habilis</i>	(i)	900 cc
b.	<i>Homo erectus</i>	(ii)	1600 cc
c.	Neanderthal man	(iii)	650-800 cc
d.	Cro-Magnon man	(iv)	1400 cc

(1) a(i), b(ii), c(iv), d(iii)

(2) a(ii), b(iii), c(i), d(iv)

(3) a(iii), b(ii), c(i), d(iv)

(4) a(iii), b(i), c(iv), d(ii)

106. Read the statements A and B and choose the **correct** option w.r.t human females.

Statement A : Uterus is supported by ligaments attached to the pectoral region of body wall.

Statement B : The shape of the uterus is like an upright pear in its anatomical position.

(1) Both statements A and B are correct and B is the correct explanation of A.

(2) Both statements A and B are correct but B is not the correct explanation of A.

(3) Statement A is correct but Statement B is incorrect.

(4) Both Statement A and B are incorrect.

107. An American company, Eli Lilly produced A and B chains of human insulin separately, which were then linked by _____ to form mature insulin. Choose the option that fills the blank **correctly**.

(1) Ionic bonds

(2) Phosphodiester bonds

(3) Hydrogen bonds

(4) Disulfide bridges

108. Lung's alveoli get filled with fluid leading to severe problems in respiration in

(1) Common cold

(2) Typhoid

(3) Dysentery

(4) Pneumonia

109. The anterior portion of sperm head is covered by a cap-like structure called

(1) Nucleus

(2) Acrosome

(3) Middle piece

(4) Centriole

110. Withdrawal syndrome is observed in the body when

- (1) Drugs produce a sense of euphoria and increased energy.
- (2) Doses of drugs/alcohol is abruptly discontinued.
- (3) Biological response modifiers such as α -interferons are administered.
- (4) A person is unduly pressurised by peers.

111. Assertion (A) : All reproductive tract infections (RTI) are STDs but all STDs are not RTIs.

Reason (R) : All STIs are not infections of reproductive organs though their mode of transmission could be through sexual contact.

In the light of above statements, select the **correct** option.

- (1) Both Assertion and Reason are true and reason is the correct explanation of the assertion
- (2) Both Assertion and Reason are true but reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

112. Choose the **mismatch** w.r.t vector and disease

Vector	Disease
A. Anopheles mosquito	Malaria
B. Culex mosquito	Filariasis
C. Aedes mosquito	Dengue
D. Xenopsylla	Kala-azar

- (1) (A)
- (2) (B)
- (3) (C)
- (4) (D)

113. A : Oral pills are small doses of either progestogens or progestogen-estrogen combinations.

R : Pills have to be taken daily for 21 days starting preferably within the first 5-days of menstrual cycle.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

114. A : Cytotoxic T-cells are responsible for graft rejection.

R : Cyclosporin is used as immunosuppressant to avoid graft rejection.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

115. DNA has a A charge due to which it moves towards the B electrode of the electrophoretic chamber.

Choose the option that fills the blanks **correctly**.

	A	B
(1)	Positive	Positive
(2)	Negative	Positive
(3)	Negative	Negative
(4)	Positive	Negative

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

116. Choose the **incorrect** statement in context of estrogen.

- (1) Long term treatment with estrogen can cause endometrial hypertrophy
- (2) The source of estrogen is only Graafian follicle in a pregnant female
- (3) Estrogen causes proliferation of stratum functionale of endometrium
- (4) Estrogen leads to pre-mature closure of epiphyseal plates

117. A : For chemical evolution, life could have begun only in the absence of free oxygen.

R : Oxygen is quite reactive and would have oxidised the organic molecules that are necessary building blocks in the origin of life.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

118. Early diagnosis of diseases is possible by using all of the following techniques, **except**

- (1) PCR
- (2) ELISA
- (3) RDT
- (4) Serum analysis

119. Read the following features.

- a. Suppresses sperm motility.
- b. Promotes phagocytosis of sperms within uterus.
- c. Copper ions reduce the fertilizing capacity of sperms.
- d. Inhibits ovulation.
- e. Blocks the entry of sperms beyond cervix.

Which of the above are applicable for Multiload 375?

- (1) b & e only
- (2) b, c & d only
- (3) a, b & c only
- (4) c, d & e only

120. Select the option that **correctly** represents a palindromic sequence.

- (1) 5'-GGAATTCC-3'
3'-CCTTAAGG-5'
- (2) 5'-AATTCGG-3'
3'-TTAAGGCC-5'
- (3) 5'-GATCCGA-3'
3'-CTAGGCT-5'
- (4) 5'-ATCGAGTC-3'
3'-TAGCTCAG-5'

121. Select the **incorrect** statement w.r.t. stirred-tank bioreactor

- (1) Most commonly used bioreactor
- (2) Stirrer facilitates mixing of O₂ in bioreactor
- (3) It has foam control system
- (4) Temperature and pH cannot be controlled

122.A : Selectable marker helps in identifying and eliminating non-transformants from pool of transformants.

R : Antibiotic resistance gene for ampicillin serves as the only selectable marker while cloning in pBR322.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

123.A : Spines and thorns are an example of homoplasy in plants.

R : They resemble superficially but have evolved independently to solve the common need for protection from herbivores.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

124. Read the following statements A and B and choose the **correct** option.

Statement (A) : Lactational amenorrhoea has been reported to be effective upto a period of six months following parturition.

Statement (B) : Side effects are almost nil and failure rate is high in case of natural methods of contraception.

- (1) Both statements are correct
- (2) Both statements are incorrect
- (3) Only statement (A) is correct
- (4) Only statement (B) is correct

125. The property of cancerous cells to slough off from tumors and reach distant sites through blood is called

- (1) Metagenesis
- (2) Metastasis
- (3) Carcinogenesis
- (4) Contact inhibition

126.A : *Bacillus thuringiensis* produces insecticidal protein crystals as inactive protoxins to kill certain insects.

R : Inactive protoxin is converted into active toxin in gut of insects due to alkaline pH.

- (1) Both (A) and (R) are true and the (R) is correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) (A) is false but (R) is true

127. The copper releasing intra uterine devices primarily act as contraceptive by

- (1) Providing a latex barrier to sperm entry in vagina
- (2) Forming a cervical plug
- (3) Preventing ovulation
- (4) Suppressing sperm motility

128. The main challenge in the production of insulin using rDNA technology was

- (1) Getting insulin assembled in a mature form
- (2) Removing chain A and chain B and getting functional C-peptide
- (3) Searching for *E.coli* strain as host
- (4) Identifying vector for transgene delivery

129. In agarose gel electrophoresis, DNA fragments are separated from each other on the basis of

- (1) Charge
- (2) Size
- (3) Colour
- (4) Both (1) & (2)

130. Select the **correct** set of barrier methods of contraception.

- (1) Lippes loop, Cervical cap, Condom
- (2) CuT, Multiload 375, Lippes loop
- (3) Diaphragm, Condom, Vault
- (4) Condom, CuT, Cervical cap

131. Which of the following is not an IUCD?

- (1) Lippe's loop
- (2) CuT 200
- (3) LNG-20
- (4) Sponge

132. In human males, the urethra originates from (a) and extends through (b) and open outside via an opening called (c)

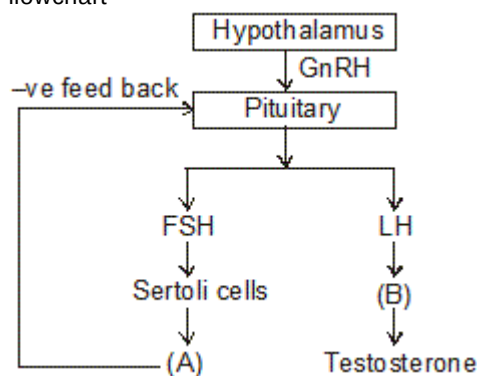
Select the option that fills the blanks **correctly**.

- (1) (a) Urinary bladder, (b) Penis, (c) Urethral meatus
- (2) (a) Penis, (b) Ejaculatory duct, (c) Urethral meatus
- (3) (a) Urinary bladder, (b) Penis, (c) Foreskin
- (4) (a) Prostate, (b) Penis, (c) Urethral meatus

133. Bt toxin of *Bacillus thuringiensis* does not harm the bacteria itself because

- (1) Production of toxin is possible only in a heterologous host
- (2) Toxin amount is negligible in the host cell
- (3) Toxin exists as inactive protoxin in the host cell
- (4) The host cell digests the toxins using its restriction endonucleases

134. Select the **correct** option w.r.t. (A) and (B) in the following flowchart



- (1) (A) – ABP, (B) – Interstitial cells
- (2) (A) – AMF, (B) – Spermatogonia
- (3) (A) – Inhibin, (B) – Spermatogonia
- (4) (A) – Inhibin, (B) – Leydig cells

135. Antigen binding site of an antibody is situated towards

- (1) N-terminal of light chain only
- (2) C-terminal of both light and heavy chains
- (3) N-terminal of heavy chain only
- (4) N-terminal of both light and heavy chains

136. A : β -galactosidase gene is used as a selectable marker in pUC 8 vector.

R : Insertion of foreign DNA in β -galactosidase gene result in inactivation of this gene. Which helps in separating transformants and non-transformants.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

137. If mammalian ovum fails to get fertilized which of the following is least likely to happen?

- (1) Corpus luteum will disintegrate
- (2) Progesterone secretion rapidly declines
- (3) Endometrium of uterus is not maintained
- (4) Increase in blood supply of uterus

138. Assertion : During cleavage in a human zygote there is marked decrease in the DNA to cytoplasmic ratio.

Reason : Cleavage is important to increase metabolism of the cell to the maximum as well as to decrease the size of nucleus within cells.

In the light of above statements, choose the **correct** answer.

- (1) Both Assertion & Reason are true and the Reason is the correct explanation of the Assertion
- (2) Both Assertion & Reason are true but the Reason is not the correct explanation of the Assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

139. Which of the following does/do induce active immunity?

- (1) Colostrum
- (2) Acellular vaccine
- (3) A natural viral infection
- (4) Both (2) & (3)

140. Select the pair of lymphoid organ, where lymphocyte differentiate into antigen-sensitive lymphocyte.

- (1) Bone marrow, spleen
- (2) Bone marrow, tonsils
- (3) Bone marrow, thymus
- (4) Peyer's patches, tonsils

141. Approximately, at the end of which month of pregnancy, eyelids separate and eyelashes are formed?

- (1) 2nd month
- (2) 3rd month
- (3) 5th month
- (4) 6th month

142. A : For tissue transplantation, MHC of donor and recipient should be well matched.

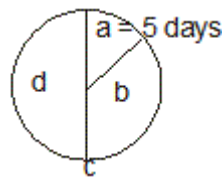
R : MHC molecules are responsible for stimulating the rejection of tissue grafts and organ transplants.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

143. In a 35 day menstrual cycle,

- a = Menstrual phase,
b = Proliferative phase
c = Day of ovulation
d = Luteal phase

if a = 5 days then, what will be the value of d and the day of ovulation?



- (1) d = 12 & the day of ovulation will be 14th day
- (2) d = 14 & the day of ovulation will be 21st day
- (3) d = 15 & the day of ovulation will be 21st day
- (4) d = 11 & the day of ovulation will be 14th day

144. Autonomously replicating circular extra-chromosomal ds DNA is called

- (1) Primer
- (2) Clone
- (3) Probe
- (4) Plasmid

145. Match the List I with List II w.r.t diseases and their causative agents.

	List I		List II
a.	Malignant malaria	(i)	<i>Epidermophyton</i>
b.	Typhoid	(ii)	<i>Plasmodium falciparum</i>
c.	Pneumonia	(iii)	<i>Haemophilus influenzae</i>
d.	Ringworms	(iv)	<i>Salmonella typhi</i>

Select the correct option.

- (1) a(iii), b(i), c(iv), d(ii)
- (2) a(ii), b(i), c(iv), d(iii)
- (3) a(ii), b(iv), c(iii), d(i)
- (4) a(i), b(iii), c(iv), d(ii)

146. A : Disturbance of Hardy Weinberg equilibrium in a population would be interpreted as resulting in evolution.
R : This principle says that allele frequencies in a population are stable and constant from generation to generation.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

147. A : Ruptured hymen is not a reliable indicator of sexual experience.

R : In some women, it can also be broken by a sudden fall or due to active participation in sports like cycling.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

148. A : In PCR specific segment of DNA is amplified.

R : Taq polymerase polymerises nucleotides in 3'-5' direction.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

149. A : In humans, testes are situated outside the abdominal cavity in scrotum to facilitate spermatogenesis.

R : This pouch like structure helps in maintaining the low temperature of the testes necessary for spermatogenesis along with the help of certain muscles.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

150. The cause of sudden increase of LH hormone in the mid cycle is

- (1) Negative feedback of estrogen on the anterior pituitary
- (2) Negative feedback of progesterone on the hypothalamus
- (3) Positive feedback of estrogen on the anterior pituitary
- (4) Positive feedback of FSH on the ovary

151. Innate immunity is accomplished by providing different types of barriers to the entry of the foreign agents into our body.

Given are certain examples for different type of barriers.

- (a) PMNL
- (b) Saliva
- (c) Macrophages
- (d) Skin

Select the correct option.

- (1) (a) and (b) are physiological barriers
- (2) (b) and (d) are physical barriers
- (3) (a) and (c) are cellular barriers
- (4) (c) and (d) are cytokine barriers

152. Which of the following restriction enzyme does **not** produce sticky ends?

- (1) *EcoRI*
- (2) *Hind III*
- (3) *Bam HI*
- (4) *Sma I*

153. Which of the following diagnostic techniques is used for the early detection of HIV?

- (1) Urine analysis
- (2) PCR
- (3) Serum analysis
- (4) Biopsy

154. Intra Uterine Transfer technique in ART involves transfer of

- (1) Sperms into uterus
- (2) Embryo with more than 8 blastomeres into uterus
- (3) Zygote into uterus
- (4) Zygote into ovary

155. By consuming golden rice we can prevent the kind of blindness which is associated with the deficiency of

- (1) Vitamin C
- (2) Vitamin A
- (3) Vitamin B
- (4) Omega 3 fatty acid

156. 'Menarche' is a term used for

- (1) First menstruation
- (2) Cessation of menstruation
- (3) Irregular menstruation
- (4) Regular heavy menstruation

157. Bollworm successfully attacks

- (1) Bt Tomato
- (2) *Bacillus thuringiensis*
- (3) Cotton plant
- (4) Bt cotton

158. Assertion (A) : IUDs are one of the most widely accepted method of contraception in India.

Reason (R) : IUDs are highly effective but their reversibility rate is very poor.

- (1) Both Assertion & Reason are true and reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

159. The immunoglobulin which is found in colostrum is

- (1) IgG
- (2) IgM
- (3) IgA
- (4) IgD

160. The fossil of this dinosaur was about 20 feet in height and had huge fearsome dagger like teeth. It could possibly be

- (1) *Ichthyosaurs*
- (2) *Tyrannosaurus*
- (3) *Brachiosaurus*
- (4) *Stegosaurus*

161. Malaria is caused by a

- (1) Helminth
- (2) Protist
- (3) Virus
- (4) Fungus

162. Conditions which are present on early Earth during chemical evolution include

- (1) High temperature, reducing atmosphere containing O_2 , NH_3 etc.
- (2) High temperature, reducing atmosphere, volcanic eruptions
- (3) Storms, low temperature, oxidising atmosphere
- (4) Oxidising atmosphere containing CH_4 , NH_3 , etc.

163. A : Lymphocytes (B/T) are useless for a person unless and until they have antigen specific receptors on them.
R : Secondary lymphatic organs are involved in differentiation of immature lymphocytes into antigen sensitive cells by adding antigen specific receptors.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

164. All of the following structures are male sex accessory ducts, **except**

- (1) Seminiferous tubules
- (2) Vasa efferentia
- (3) Rete testis
- (4) Epididymis

165.A : During electrophoresis, larger the size of DNA fragment, closer it will be to the cathode.

R : Larger fragments face more resistance as compared to smaller ones while moving through agarose matrix.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

166.During menstrual cycle, in a non-pregnant female, event unlikely to occur is

- (1) Shedding of endometrium due to reduced level of progesterone
- (2) Ovulation in mid of the cycle
- (3) Positive feedback by progesterone causing LH surge
- (4) Repair of endometrium under estrogen influence

167.Reproductive and Child Health care (RCH) programme aims to focus at all the following **except**

- (1) Creating awareness about reproduction related aspects.
- (2) Providing facilities for maintaining reproductive health.
- (3) Providing support for building a reproductively healthy society.
- (4) Providing integrated educational services, for children.

168.A novel approach for the treatment of cancer is the use of biological response modifiers called _____ which activate the immune system to destroy the tumor. Choose the option which fills the blank **correctly**.

- (1) Vinblastin
- (2) X-rays
- (3) α -interferons
- (4) Oncogenes

169.Choose the incorrect match:

	Column I (Disease)	Column II (Caused by)
(1)	Chlamydiasis	Bacterium
(2)	Gonorrhoea	Bacterium
(3)	Trichomoniasis	Ciliated protozoan
(4)	AIDS	Virus

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

170.The applications of biotechnology include all of the following, **except**

- (1) Agro-chemical based farming
- (2) Development of pest resistant plants
- (3) Bioremediation
- (4) Therapeutics

171.Consider a cat population of 100, in which 84 are black and 16 are white. Black trait is dominant over white. What will be the number of heterozygotes if this population is in Hardy-Weinberg equilibrium?

- (1) 16
- (2) 36
- (3) 48
- (4) 84

172.PCR technique using *Taq* enables directly the a of billions of copies of an original piece of b in a tube within minutes. Select the option which fills the blanks (a and b) **correctly**?

- (1) a-Synthesis, b-RNA
- (2) a-Amplification, b-DNA
- (3) a-Synthesis, b-Protein
- (4) a-Amplification, b-Protein

173.Anteater is a placental mammal while Numbat is an Australian marsupial but both exhibit structural similarities as well as lead similar ways of life. This phenomenon of two different organisms converging towards similarity is called

- (1) Adaptive radiation
- (2) Co-evolution
- (3) Convergent evolution
- (4) Divergent evolution

174.Read the following statements carefully

Statement-A : The native (wild type) *E.coli* cells do not carry resistance against any antibiotics

Statement-B : Lac Z gene in pBR322 is considered as useful selectable marker for *E.coli*

Choose the **correct** option

- (1) Both statements are incorrect
- (2) Both statements are correct
- (3) Only statement A is correct
- (4) Only statement B is correct

175. Match column I with column II and choose the **correct** option.

	Column I		Column II
a.	Menstrual phase	(i)	Secretory phase
b.	Follicular phase	(ii)	Release of ovum
c.	Ovulatory phase	(iii)	Proliferative phase
d.	Luteal phase	(iv)	Bleeding phase

- (1) a(i), b(iii), c(ii), d(iv)
 (2) a(i), b(ii), c(iii), d(iv)
 (3) a(iv), b(iii), c(ii), d(i)
 (4) a(ii), b(iii), c(i), d(iv)

176. Which of the following originated from the early reptiles?

- (i) Modern day reptiles
 (ii) Mammals
 (iii) Birds

- (1) (i) & (ii) only
 (2) (i), (ii) & (iii)
 (3) (i) only
 (4) (iii) only

177. Complete the analogy

Tasmanian tiger cat : Bobcat :
 Spotted cuscus : ____

- (1) Anteater
 (2) Wolf
 (3) Lemur
 (4) Flying squirrel

178. Secretion of which of the following glands helps in the lubrication of penis?

- (1) Prostate gland
 (2) Sweat gland
 (3) Seminal vesicles
 (4) Bulbourethral glands

179.A : In chronic smokers, blood pressure and heart rate increases.

R : Nicotine stimulates adrenal gland to release adrenaline and non-adrenaline into blood circulation.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
 (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
 (3) Assertion is true statement but Reason is false
 (4) Both Assertion and Reason are false statements

180.Assertion (A) : More and more children in metro cities of India suffer from allergies due to sensitivity to the environment.

Reason (R) : It is generally due to protected environment provided early in life.

- (1) Both (A) and (R) are true and (R) is the correct explanation of the (A)
 (2) Both (A) and (R) are true but (R) is not the correct explanation of the (A)
 (3) (A) is true statement but (R) is false
 (4) Both (A) and (R) are false statements